Deep Learning and Applications

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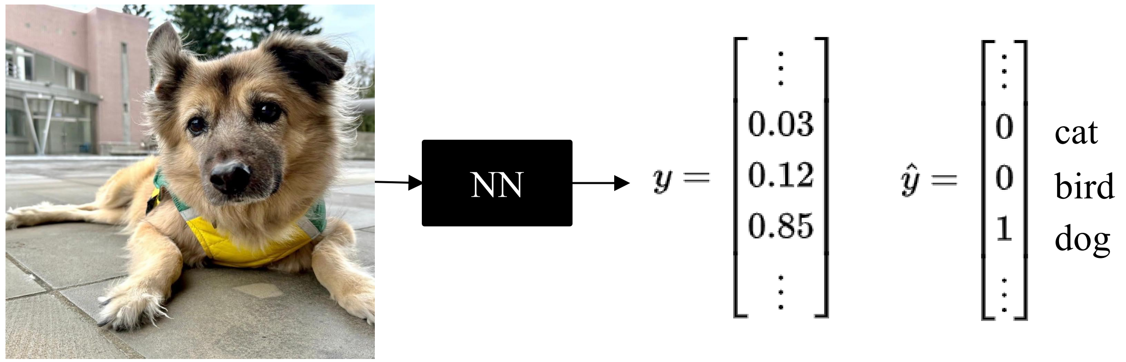
Challenges Faced by Deep Neural Networks (DNNs)

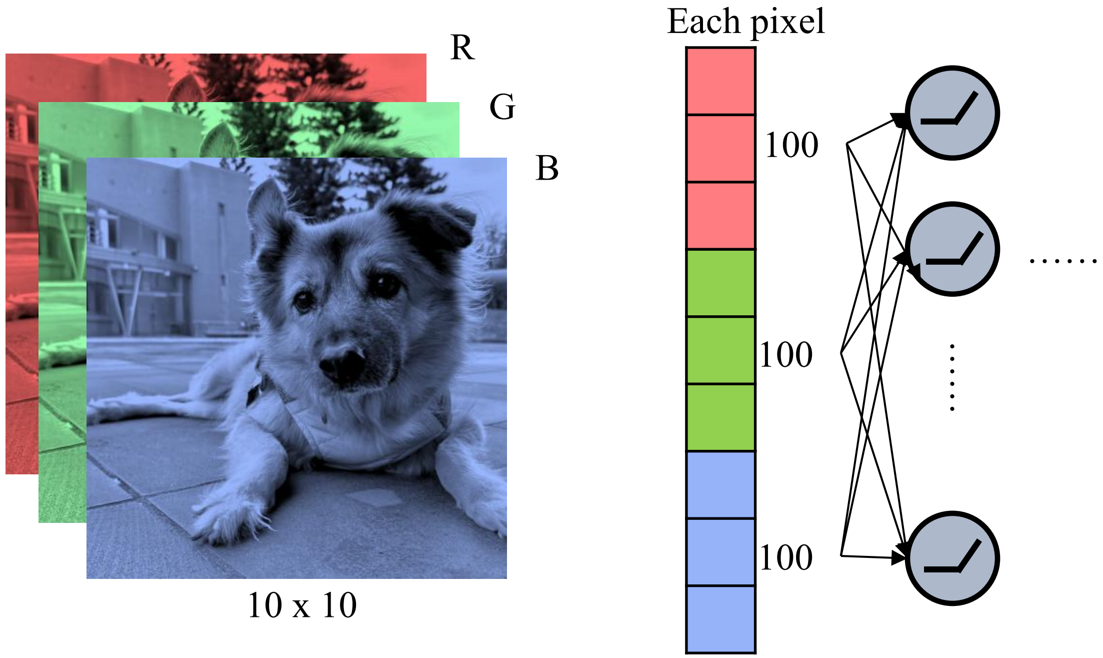
- Problem Overview
  1. Traditional DNNs use **fully connected layers**
  2. Every output unit interacts with **all input units**
- This leads to:
  1. **Massive number of parameters**
  2. **High computational cost**
  3. Poor statistical efficiency

     When the training data is limited, the model may overfit—memorizing the training examples rather than learning generalizable patterns.
  4. Inflexibility with input size

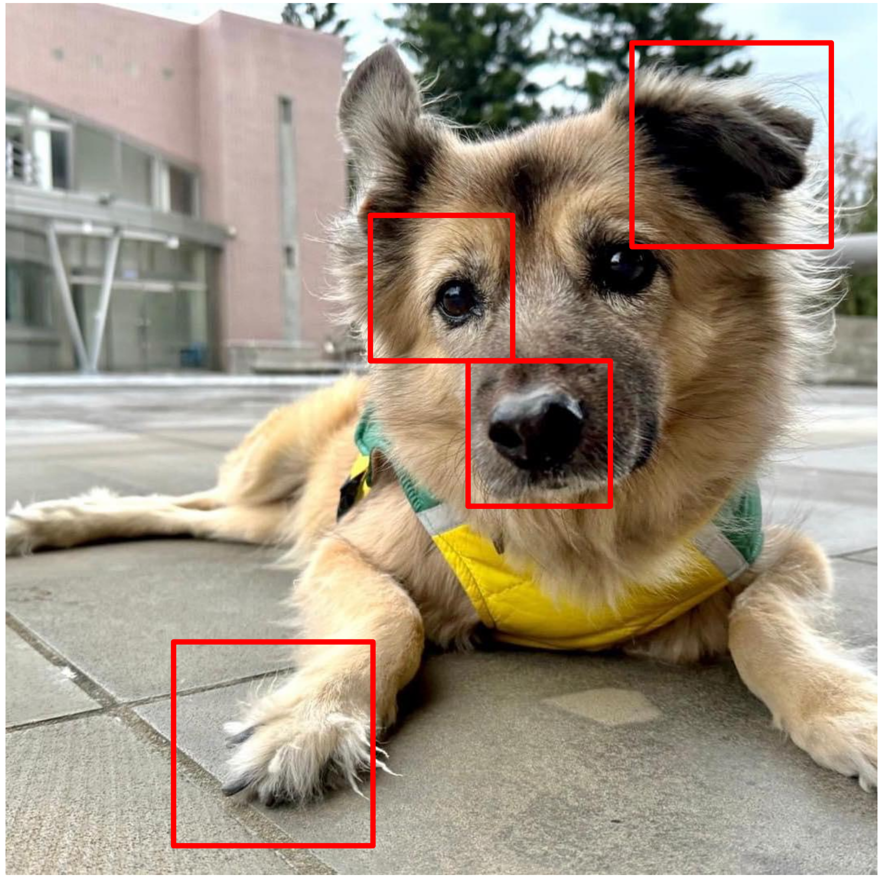
     $x \in \mathbb{R}^{100}$ ➡ $W \in \mathbb{R}^{m \times 100}$

     → The **input dimension must match the weight matrix**





Why CNN? (Convolutional Neural Networks)



- The patterns are smaller than the entire image.
- The same patterns appear in different regions.

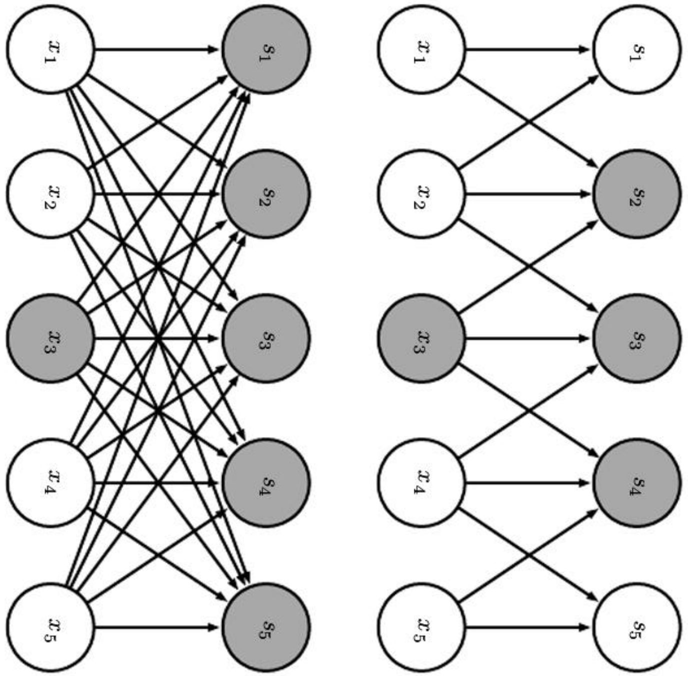
→ Can we tie the weights for certain groups and prune several weights?

Parameter Sharing?

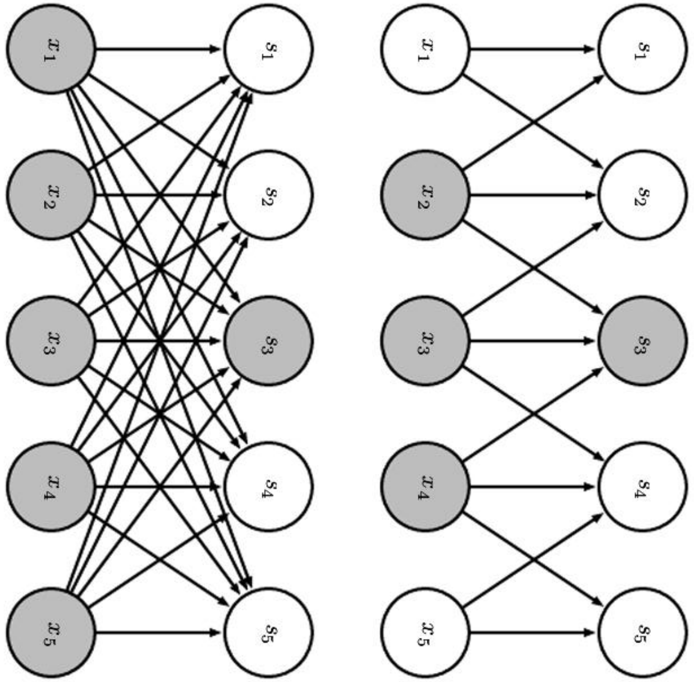
The set of neurons may be with the same set of parameter.

CNN’s Key Ideas (9.3)

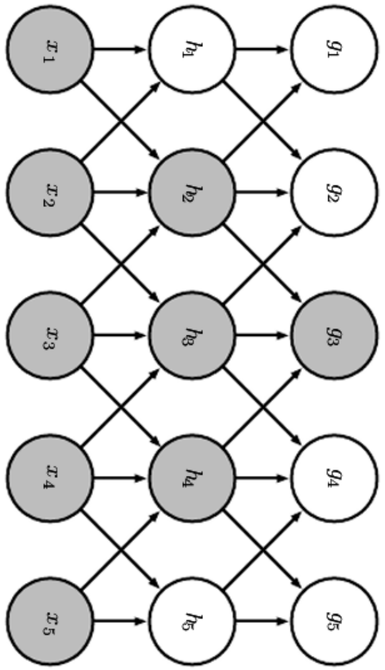
- Sparse Connectivity
  1. Each neuron connects only to a **local region** of the input, rather than all input units (as in fully connected layers).
  2. Reduces the number of parameters → **less memory** and **faster computation**.
  3. Focuses on meaningful **local features**



convolution with a kernel of width 3



receptive field of s_3



receptive field of g_3

CNN's Key Ideas (Convolutional Neural Networks)

- Parameter Sharing
  1. The **same set of weights** (kernel) is applied across **all spatial locations**.
  2. Each parameter is **reused** many times → **fewer parameters**, **better generalization**.
  3. Greatly improves **statistical efficiency**.
- Equivariance
  1. Convolution is **equivariant to translation**:
     if the input shifts, the output shifts in the same way.
  2. Ensures the network preserves **spatial relationships**.
  3. Foundation for capturing **structure** in images.